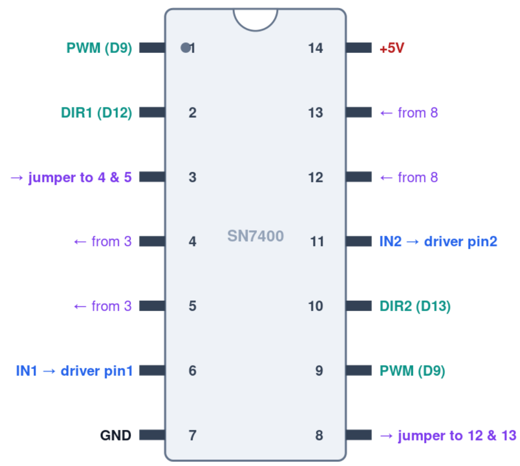
MSE 312 — How to wire the SN7400 NAND gates

Two configurations for driving the TB6568KQ inputs IN1/IN2. Both use one SN7400 quad NAND. A NAND with its two inputs tied together acts as an inverter.

Config A — 4 gates, 3 MCU pins (IN1 = PWM·DIR1, IN2 = PWM·DIR2)

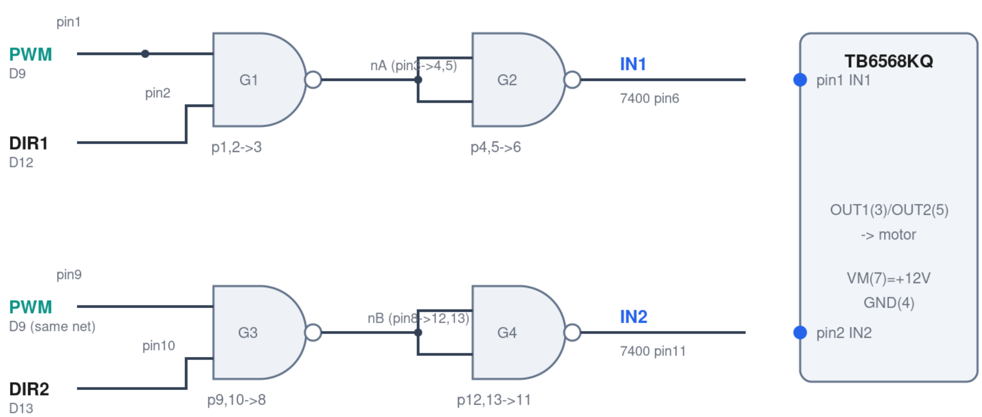
SN7400 pin map — 4-gate config (D9 PWM · D12 DIR1 · D13 DIR2)

Top view, notch up. IN1 = PWM·DIR1, IN2 = PWM·DIR2 (each AND = NAND + inverter)



PWM (D9) ties to BOTH pin1 and pin9 · all 4 gates used · pin14=+5V, pin7=GND

One SN7400 to generate IN1 / IN2 (IN = PWM AND DIR)



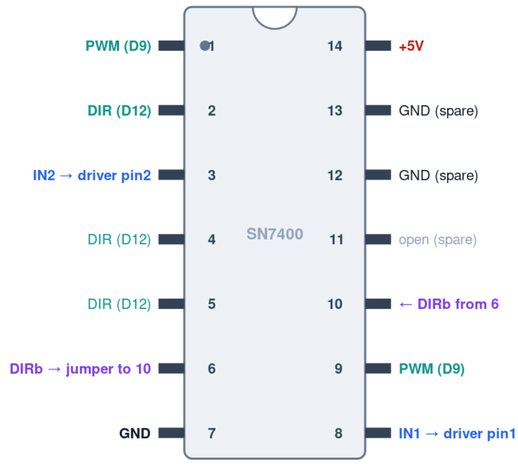
SN7400: pin14=+5V, pin7=GND · all grounds common (Arduino + supply + TB6568KQ)

From	To	Purpose
Arduino D9 (PWM)	SN7400 pin 1 AND pin 9	PWM to both NANDs
Arduino D12 (DIR1)	SN7400 pin 2	forward bit
Arduino D13 (DIR2)	SN7400 pin 10	reverse bit
SN7400 pin 3	SN7400 pins 4 & 5	gate1 out into gate2 (inverter)
SN7400 pin 8	SN7400 pins 12 & 13	gate3 out into gate4 (inverter)
SN7400 pin 6	TB6568KQ pin 1 (IN1)	IN1
SN7400 pin 11	TB6568KQ pin 2 (IN2)	IN2
SN7400 pin 14 / pin 7	+5V / GND	logic power

Config B — 3 gates, 2 MCU pins (IN2 = NAND(PWM,DIR), IN1 = NAND(PWM, NOT DIR))

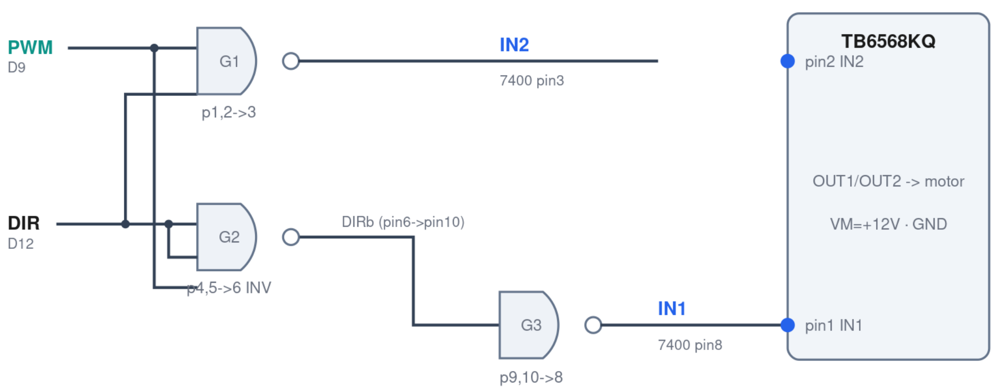
SN7400 pin map — 3-gate config (D9 PWM · D12 DIR)

Top view, notch up. IN2 = NAND(PWM,DIR), DIRb = NOT DIR, IN1 = NAND(PWM,DIRb)



PWM (D9) → pin1 & pin9 · DIR (D12) → pin2, 4 & 5 · spare gate 4 inputs (12,13) tied to GND

3 NAND gates · 2 pins -> IN1 / IN2



Spare gate 4: tie pins 12 & 13 to GND · 7400 pin14=+5V, pin7=GND

From	To	Purpose
Arduino D9 (PWM)	SN7400 pin 1 AND pin 9	PWM to both NANDs
Arduino D12 (DIR)	SN7400 pins 2, 4 & 5	direction + inverter input
SN7400 pin 6 (DIRb)	SN7400 pin 10	inverted DIR into gate 3
SN7400 pin 3	TB6568KQ pin 2 (IN2)	IN2
SN7400 pin 8	TB6568KQ pin 1 (IN1)	IN1
SN7400 pins 12 & 13	GND	tie off spare gate 4
SN7400 pin 14 / pin 7	+5V / GND	logic power

Common to both: the SN7400 runs on 5 V (pin14=+5V, pin7=GND) tied to the Arduino 5V/GND. Never leave an unused NAND input floating — in Config B the spare gate-4 inputs (pins 12,13) go to GND. IN1/IN2 always land on TB6568KQ pins 1 and 2 respectively; OUT1/OUT2 (pins 3,5) go to the motor.